

AVNISH SINGH

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B.Tech CSE, graduating May 2027 | Eligible to work in India | Open to relocation

SUMMARY

Final-year CS student who ships deployed systems and measures them honestly: a multi-agent crisis platform where a deterministic rule engine gates every LLM output, a deployed multimodal medical image classifier, and a benchmark for silent omissions in coding-agent patches — headline result contradicting my starting thesis, shipped as a CI gate, with a companion preprint pending arXiv. Looking for ML engineering work where models have to survive contact with production.

PROJECTS

- AegisOps AI — Multi-Agent Crisis Coordination Platform2026

Python, agentic AI, LLM orchestration, FastAPI, SQLite, Docker, pytest | github.com/Avnish1505/aegisops-ai | [engineering blog](#)

 - Four specialised agents (perception, resource allocation, comms, safety audit) reason over live telemetry, but nothing reaches the operator until a deterministic rule engine clears it — the LLM proposes, the gate decides.
 - Reviewing AI-generated diffs for architectural correctness, not just syntax, caught my coding assistant scaffolding the safety gate twice without wiring it; every session now ends with pasted test output. 121 tests passing across the rule engine, governance, and LLM layers.
 - Every agent decision writes a reasoning trace and confidence score to SQLite via ORM, so a run can be reconstructed after the fact instead of taken on trust.
- OmitBench — Benchmark for Silent Omissions in Coding-Agent Patches2026

Python, AST/program analysis, mutation testing, LLM-as-judge eval, GitHub Actions | github.com/Avnish1505/Omitbench | [engineering blog](#)

 - Built a reproducible data pipeline extracting merged commits from 8 Python repositories into a labelled corpus across three omission classes — 310 instances, 2,184 requirements scored per detector, 27.9% base rate, cluster-bootstrap CIs at instance level.
 - Headline result contradicts my starting thesis: the AST detector beats a no-AST grep baseline (Δ MCC +0.128) and a reasoning LLM judge (+0.111) but loses to a mid-tier judge (-0.202, CI excludes zero); MCC pre-committed over F1 — flag-everything scores F1 0.44 at MCC 0.000.
 - Removed the benchmark’s hidden oracle by extracting requirements from issue text instead of the gold patch — MCC collapsed from +0.499 to -0.935, worse than random; the \$0.06 ablation showed extraction, not detection, is the hard problem, killing the free-text design.
 - Shipped as a GitHub Action PR gate, verified live in 8s — p95 26.5ms per instance, 132 tests in CI. Shipped the lower-MCC detector on purpose: a gate that cries wolf gets switched off, so precision was the ship criterion, not MCC. It cleared the 0.80 bar at 2.6x lower FPR; two detectors missing a pre-agreed bar were retired.
- Cancer Fusion AI — Multimodal Skin Lesion Classifier2026

PyTorch, ResNet50, computer vision, Grad-CAM, FastAPI, React, Vercel | github.com/Avnish1505/cancer-fusion-ai

 - Fused ResNet50 image features with patient metadata via an MLP head on HAM10000 (10,015 dermoscopy images, 7 classes) — 0.726 macro-F1, reported over accuracy because one class is two-thirds of the data.
 - Melanoma recall 0.672 at precision 0.390: a missed melanoma costs more than an unnecessary referral, so this is a triage filter, not a diagnostic. Same reasoning drove WeightedRandomSampler over shuffling.
 - Split data at lesion level, not image level: the same lesion on both sides of the boundary would have inflated validation accuracy without improving the model.
 - Profiled inference per phase — an unconditional Grad-CAM backward pass dominated request compute; moved it behind an opt-in explain flag, median latency 100ms to 31ms, predictions verified bit-identical against the pre-change build.

TECHNICAL SKILLS

LLM / Agentic Systems: agentic AI, AI agents, agentic workflows, human-in-the-loop gating, multi-agent orchestration, LLM application development, deterministic safety gating, reasoning-trace logging, prompt engineering, AI-assisted coding tools (Claude, Copilot), AI-generated code review

Evaluation / Benchmarking: LLM evaluation, eval harness design, LLM-as-judge evaluation, silent-failure and omission detection, benchmark design, static analysis, AST analysis, mutation testing, MCC and bootstrap confidence intervals, leakage guards, reproducible pipelines

ML / Deep Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, computer vision, CNNs, image classification, transfer learning, Grad-CAM, class imbalance handling, model evaluation, model serving, inference optimization, latency profiling

Cloud / Data / Software Engineering: AWS, SQL, SQLite + ORM, reproducible data pipelines, data validation, Python, FastAPI, REST APIs, backend development, production deployment, Docker, Linux, React, pytest, unit testing, GitHub Actions, CI/CD, Git, data structures and algorithms

Also: NumPy, Pandas, Streamlit, Vercel, technical writing

RESEARCH

- Detecting Silent Implementation Integrity Failures in AI-Generated Code: A Static Analysis Approach — Preprint, arXiv submission pending (2026) | [preprint](#)

 - Built a deterministic AST-based static analyzer detecting three structural signatures of unwired safety logic — MCC 0.389 (precision 0.556, recall 0.833) on a 15-scenario benchmark, against a string-presence baseline whose precision and MCC are mathematically undefined (single-class predictor).
 - Reported all five classification errors (4 false positives, 1 false negative) against limitations documented before the benchmark was run; the false negative demonstrates that call-set presence is not execution ordering.
- Emergence of Artificial Intelligence in Law and Legal Technology — ADG 2026 International Conference | [paper](#)

 - Co-authored a peer-reviewed comparison of AI governance regimes across the EU, US, and India, covering algorithmic bias and predictive justice. Wrote the comparative analysis and technical sections.

EDUCATION

B.Tech, Computer Science and Engineering

2023 – 2027 (expected May 2027)

Babu Banarasi Das University, Lucknow | CGPA 7.55 / 10

ADDITIONAL

- 140 LeetCode problems solved — arrays, strings, hash tables, dynamic programming, DFS. leetcode.com/u/Avnish1506
- Shortlisted for Anthropic’s Claude for Open Source Program on code quality. First place, college chess tournament 2025 (Chess.com rapid 1720).